

Result Analysis

Model	Acc.	Prec.	Rec.	F1-score
BanglaBERT	97.9%	98.2%	99.3%	98.8%
XLM-R	97.3%	97.6%	99.3%	98.4%
mBERT	87.3%	87.7%	86.7%	87.2%
Gemma-2B	84.4%	71.2%	84.4%	77.2%
Gemini	73.0%	75.6%	73.0%	63.9%
Bangla-s1k-llama-3.2-3B	97.17%	97.39%	96.85%	97.09%
culturax-base-3b-seqcls	97.21%	97.27%	97.02%	97.14%
BanglaLLama-3.2-3b-alpaca	97.21%	97.33%	96.97%	97.14%
Bangla-s1k-qwen-2.5-3B	97.36%	97.48%	97.13%	97.29%
Majority vote (ensemble)	97.50%	97.67%	97.24%	97.44%
Soft vote (ensemble)	97.62%	97.70%	97.43%	97.56%
Weighted soft vote (ensemble)	97.62%	97.70%	97.43%	97.56%
Stacking (ensemble)	97.39%	97.54%	97.15%	97.33%
Bangla-s1k-qwen-2.5-3B (Prompt)	78.67%	73.35%	78.67%	75.70%
Bangla-s1k-llama-3.2-3B (Prompt)	78.84%	73.31%	78.84%	75.74%
BanglaLLama-3.2-3b-alpaca (Prompt)	78.90%	73.34%	78.90%	75.77%
culturax-base-3b (Prompt)	78.90%	73.34%	78.90%	75.77%